

ogy. It is not just waste, it is a cocktail of some very toxic substances such as mercury, lead, cadmium, arsenic, beryllium and brominated flame retardants. Toxic materials in electronics can cause cancer, reproductive disorders, endocrine disruptions and many other health-related problems if this waste stream is not properly managed.

Uncontrolled discarding or inappropriate waste management/recycling generates significant hazardous emissions, with a severe impact on health and environment. This is the biggest challenge, in particular, in developing and transition countries, where backyard recycling with open-sky incineration is done to recover copper from cables, cyanide leaching to release gold, acid cooking of circuit boards and the like.

The poisonous residues being dumped in drains and rivers is a common practice. Most environmental damage and health impacts related to e-waste arise from improper

Health hazards caused due to the presence of toxic materials found in e-waste

E-waste contains some very toxic substances whose effects on humans are devastating as detailed below:

- Lead (Pb) is toxic to kidneys and its accumulation in the body eventually affects blood, nervous and reproductive systems. Children's mental development can be impaired by even low-level exposure to lead.
- Mercury (Hg) is known to harm developing fetuses and is passed through mother's milk to newborns. It can cause brain and kidney damage in adults, too.
- Cadmium is a carcinogen and long-term exposure to it can lead to kidney and bone damage.
- Beryllium causes acute or chronic beryllium disease, a deadly ailment affecting the lungs.
- Brominated flame retardants have been linked to fetal damage and thyroid problems.
- Incineration of PVC produces dioxins, some of the most hazardous carcinogens.
- Hexavalent chromium damages kidney, liver and DNA. Asthmatic bronchitis has been also linked to this substance.

collection and treatment approaches.

Refer box above for the various health hazards that are caused due to different toxic materials in e-waste.

Dangerous effects of improper e-waste disposal is not limited to humans; it hurts the planet and all its inhabitants including animals, birds, sea mammals and aquamarine life. Virtually every living organism is

impacted by the dangerous chemical pollutants that are released into the atmosphere and underground water when e-waste is improperly disposed.

Polluted soil and air contaminates vegetables, crops, flowers, grasses, herbs, plants, saplings, shrubs and trees. Everyone and everything exposed to toxic pollutants in e-waste is at risk. **EFY**

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